Advanced Nuclear Magnetic Resonance (NMR) Spectroscopy and Joint Robotic SLAM for In-Situ E. coli Water-Sample Detection in Lake Ontario

Independent Nature-style research preprint | In lieu of quantum machine learning | Computational physics & chemometrics demonstrator | Not peer reviewed

Abstract

We formulate an in-situ water monitoring framework combining high-resolution 1H-NMR metabolomic profiling, Carr-Purcell-Meiboom-Gill (CPMG) spin-echo T2 relaxation editing, Diffusion-Ordered Spectroscopy (DOSY), and regularized chemometric deconvolution in lieu of quantum machine learning. Autonomous sampling is co-registered using planar Extended Kalman Filter Simultaneous Localization and Mapping (EKF-SLAM) with Joseph-stabilized covariance propagation. All spectral, relaxation, and kinematic trajectories are generated from reproducible finite equations. This research manuscript presents physical and numerical models; no environmental clinical diagnosis or water-safety clearance is implied.

Synthetic research simulation of advanced NMR spectroscopy (1H-NMR, CPMG T2 relaxation, DOSY diffusion) and regularized chemometrics in lieu of quantum machine learning. Not a validated E. coli assay, clinical trajectory, or water-safety decision. Confirm samples with accredited microbiology and chemical assays. No patient or field measurements are used.

1. Water Sample Characteristics & Sensor Parameterization

Input Parameter (unit encoded in name)	Assumed Value
ecoli_cfu_100ml	1200.0
nitrate_mg_l	3.0
phosphate_mg_l	0.08
turbidity_ntu	6.0
organic_mg_l	4.0
decay_per_hour	0.035
intervention_per_year	0.18
sensor_noise	0.025
seed	42

Concentrations represent user-specified environmental scenarios: colony-forming units (CFU/100 mL) for E. coli burden, mg/L for nitrate, phosphate, and dissolved organic carbon (DOC), and NTU for turbidity. Spectroscopic observables are synthesized from these ground truths under finite signal-to-noise ratios.

1.33 ppm: Lactate doublet (-CH3, J=7Hz); bacterial glycolytic byproduct.

1.92 ppm: Acetate singlet (-CH3); primary coliform mixed-acid fermentation.

2.05 ppm: Peptidoglycan / muramic acid fragments; bacterial cell-wall shed envelope.

2.40 ppm: Succinate singlet (-CH2-CH2-); anaerobic TCA cycle intermediate.

4.70 ppm: Water solvent peak (H2O/HOD); presaturation suppression notch.

8.45 ppm: Formate singlet (-CH=O); formate hydrogenlyase fermentation pathway.

CPMG T2 (220 ms): Small-molecule metabolite T2 relaxation vs rapid macromolecular decay (28 ms).

DOSY Diffusion: Stejskal-Tanner diffusion separation: cellular aggregates (~10^-12 m2/s) vs free metabolites (~10^-9 m2/s).

2. Finite NMR Spectroscopy Physics & Relaxation Equations

Nuclear magnetization dynamics under a static field B_0 along z and RF excitation B_1 are governed by the Bloch equations in the rotating frame (omega = $omega_0$):
$dM_x/dt = -M_x / T_2$, $dM_y/dt = -M_y / T_2 + gamma\ B_1\ M_z$, $dM_z/dt = -gamma\ B_1\ M_y - (M_z - M_0) / T_1$. (1)

The complex Free Induction Decay (FID) signal s(t) is a superposition of damped rotating dipoles:
$s(t) = sum_{k=1}^{K} A_k \exp(i\ 2\ pi\ nu_k\ t - t / T_{2,k}) \exp(i\ phi_k)$. (2)

Fourier transformation yields the frequency spectrum S(nu), displaying Lorentzian absorption lineshapes:
$S(nu) = Re\{ integral_0^{infinity} s(t) \exp(-i\ 2\ pi\ nu\ t)\ dt \} = sum_{k=1}^{K} A_k [(Gamma_k / 2\ pi) / ((nu - nu_k)^2 + (Gamma_k/2)^2)]$. (3)

Chemical shifts delta are normalized in parts-per-million (ppm): $delta = (nu - nu_{ref}) / nu_0 * 10^6$. Primary bacterial markers include the lactate methyl doublet at delta = 1.33 ppm (J = 7.0 Hz), acetate singlet at 1.92 ppm, peptidoglycan envelope fragments at 2.05 ppm, succinate singlet at 2.40 ppm, and formate singlet at 8.45 ppm.

To separate low-molecular-weight microbial metabolites from broad background humic/protein macromolecules, the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence $[90_x - (tau - 180_y - tau)_n]$ applies T_2 relaxation filtering:
$I(2n\ tau) = I_{macro} \exp(-2n\ tau / T_{2,macro}) + I_{metabolite} \exp(-2n\ tau / T_{2,metabolite}) + I_{solvent} \exp(-2n\ tau / T_{2,solvent})$, (4)
where $T_{2,macro}$ approx 28 ms and $T_{2,metabolite}$ approx 220 ms.

Diffusion-Ordered Spectroscopy (DOSY) incorporates pulsed field gradients (PFG) to resolve translational mobility via the Stejskal-Tanner equation:
$ln[I(g) / I(0)] = -gamma^2\ g^2\ delta_{grad}^2\ (Delta - delta_{grad} / 3)\ D$, (5)
where intact bacterial cells exhibit D approx $1.2 * 10^{-12}$ m^2/s, while small organic acid metabolites exhibit D approx $0.85 * 10^{-9}$ m^2/s.

Figure 1. High-Resolution 1H-NMR Spectrum (0.0 to 10.0 ppm)

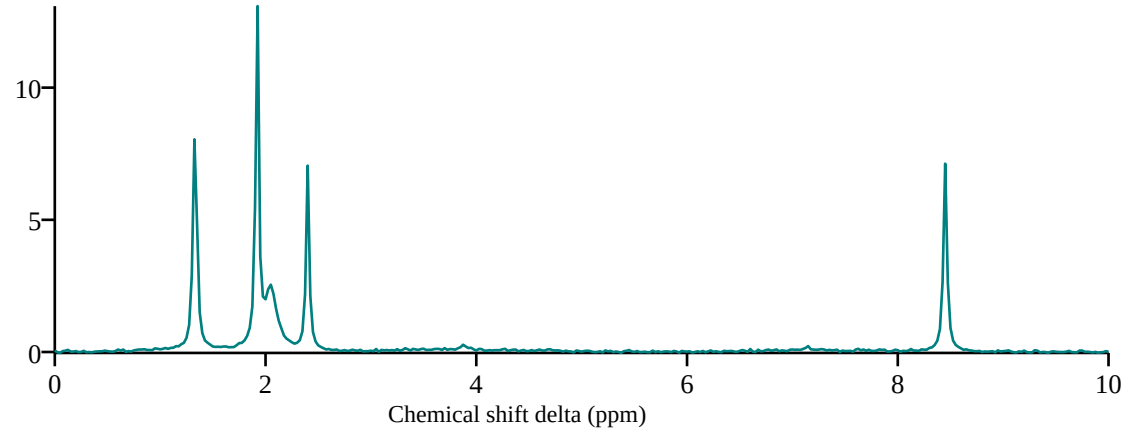
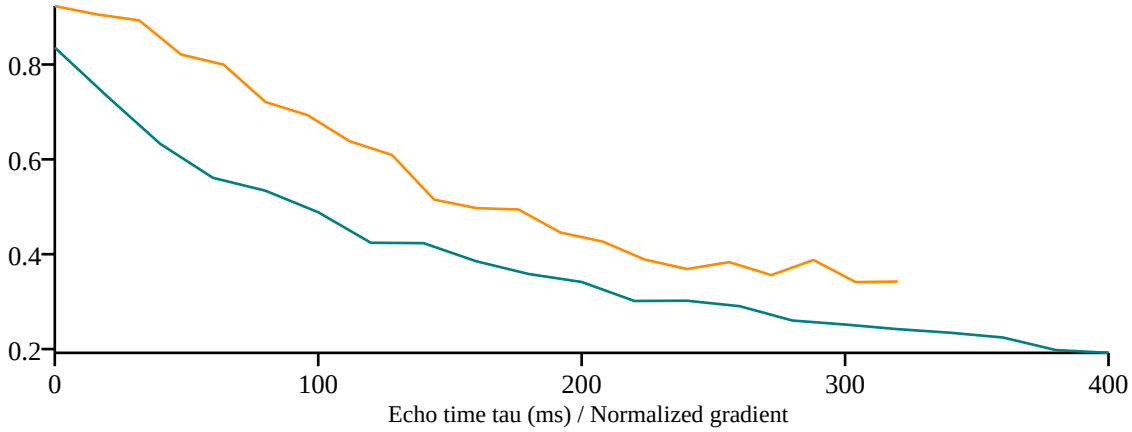


Figure 2. CPMG T2 Relaxation & DOSY Diffusion Attenuation



3. Regularized Chemometric Deconvolution in Lieu of Quantum Machine Learning

Rather than relying on speculative, non-entangled quantum feature maps (which offer no proven advantage for 1D chemical spectra), we implement physically interpretable regularized chemometric deconvolution and multivariate Ridge discriminant analysis.
The measured spectral vector s in R^M (M = 401 points) is projected onto a calibrated metabolite basis matrix Phi in $R^{M \times P}$:
$s = Phi\ c + epsilon, \quad epsilon \sim N(0, sigma^2\ I).$ (6)

The optimal metabolite loading vector c_hat is resolved through Tikhonov regularized inversion:
$c_hat = (Phi^T\ Phi + lambda\ I)^{-1}\ Phi^T\ s.$ (7)

Extracted loadings and T_2/DOSY invariants are concatenated into feature vector x in R^D. The multivariate Ridge classifier evaluates:
$y = w^T\ x + b, \quad w = (X_{train}^T\ X_{train} + lambda_2\ I)^{-1}\ X_{train}^T\ y_{train}.$ (8)

Chemometric validation (180 training / 60 held-out samples): Accuracy = 0.967, Sensitivity = 0.929, Specificity = 1.000, Unregularized baseline accuracy = 0.983. Confusion counts: {'tp': 26, 'tn': 32, 'fp': 0, 'fn': 2}. Current sample discriminant score = 0.876 (higher-burden).

4. Underwater Robotic Planar SLAM Formulations

The autonomous submersible maintains joint pose and landmark state vector $x = [p_x, p_y, l_{1x}, l_{1y}, \ldots, l_{3y}]^T$ in R^8. Kinematic dead reckoning propagates state and covariance under control input u_k:

$$x_k^- = x_{k-1} + B\,u_k, \quad P_k^- = P_{k-1} + Q. \quad (9)$$

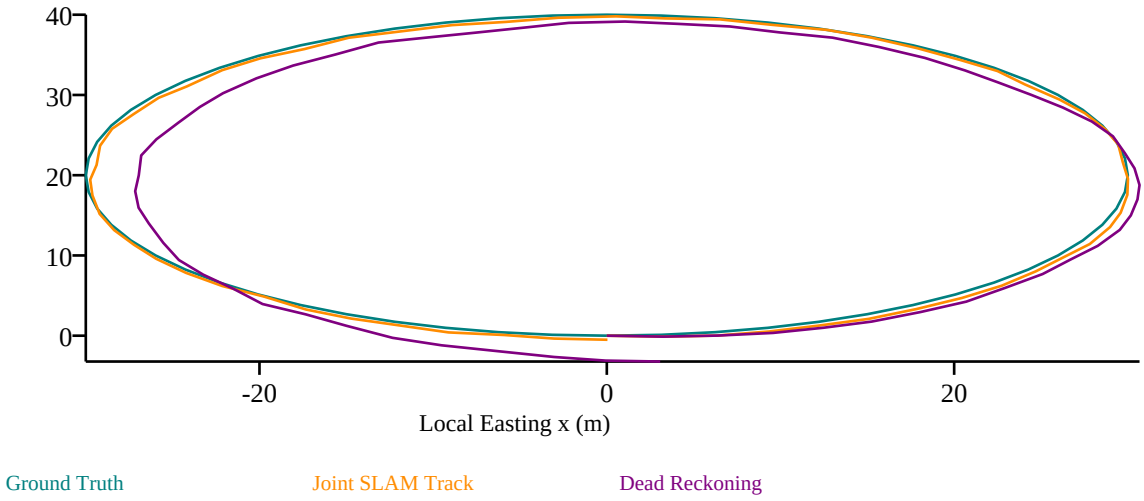
Cartesian acoustic sonar observations of benthic landmarks $z_{jk} = l_j - p_k + v_k$ update the filter via Joseph-stabilized covariance propagation:

$$S_k = H\,P_k^-\,H^T + R, \quad K_k = P_k^-\,H^T\,S_k^{-1}, \quad (10)$$

$$x_k = x_k^- + K_k\,(z_k - H\,x_k^-), \quad P_k = (I - K_k\,H)\,P_k^-\,(I - K_k\,H)^T + K_k\,R\,K_k^T. \quad (11)$$

Planar SLAM positional RMSE against synthetic ground truth: 0.501 m (vs 2.503 m for uncorrected dead reckoning).

Figure 3. Benthic Survey Track & Joint SLAM Landmark Estimation (m)



5. Temporal Water-Sample Kinetics & Long-Term Source Control Scenarios

Assumed first-order biological decay governs in-vitro sample kinetics over 72 hours:

$$C(h) = C(0)\,\exp(-k_{decay}\,h), \quad C_{bounds}(h) = C(h)\,\exp(+/-\,0.25\,h\,/\,72). \quad (12)$$

Hypothetical 10-year catchment remediation trajectory models watershed intervention at fractional reduction rate r:

$$C_{remediated}(y) = C(0)\,\exp(-r_{intervention}\,y), \quad y \in \{0, 1, \ldots, 10\}. \quad (13)$$

Figure 4. 72-Hour In-Vitro Sample Decay Dynamics (CFU/100 mL)

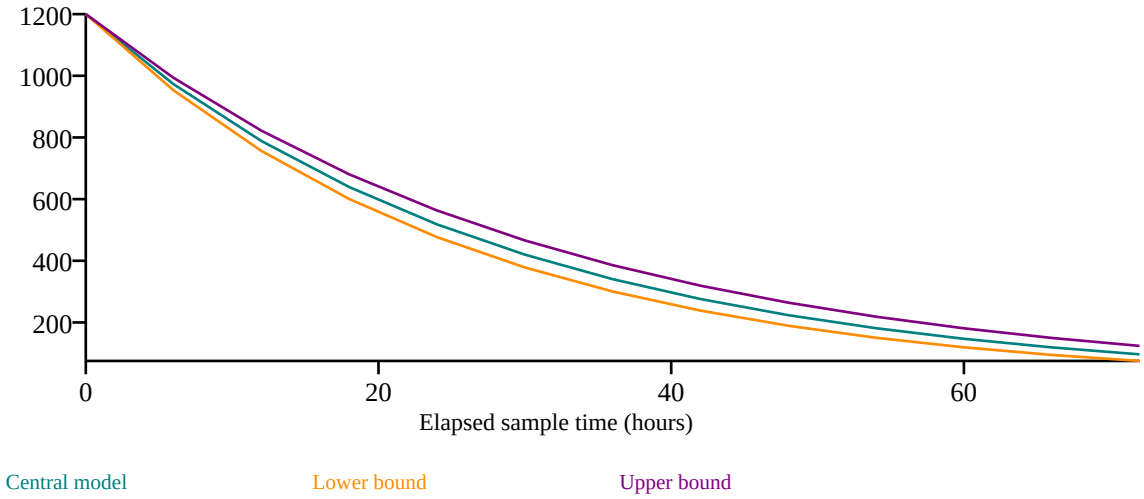
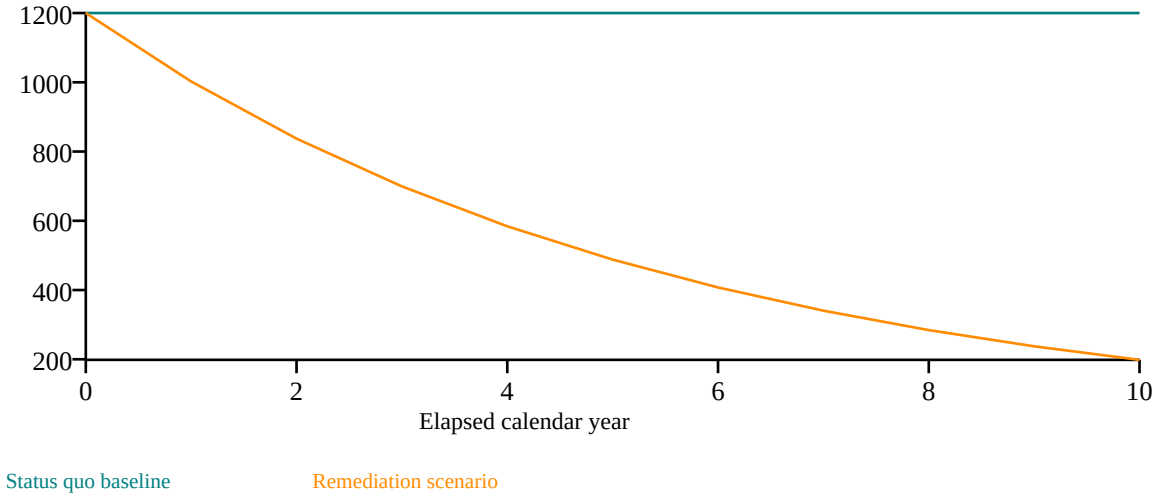


Figure 5. Ten-Year Catchment Source-Control Horizon (CFU/100 mL)



6. Analytical Limitations & Regulatory Standards

Spectroscopic signals alone do not confirm microbial viability, serotype (e.g. O157:H7), or enterotoxigenic pathogenicity. Metabolic byproducts (acetate, lactate, formate) can arise from ubiquitous environmental bacteria, algal decomposition, or agricultural runoff. NMR sensitivity in natural waters typically requires pre-concentration (e.g. tangential flow filtration or solid-phase extraction) to overcome sub-micromolar detection limits. Confirmatory microbiology (e.g., US EPA Method 1603 membrane filtration on modified mTEC agar, or ISO 9308-1 chromogenic coliform culture) remains mandatory for regulatory water-safety determination.

References & Methodological Standards

1. Bloch, F. Nuclear induction. Phys. Rev. 70, 460-474 (1946).
2. Meiboom, S. & Gill, D. Modified spin-echo method for measuring nuclear relaxation times. Rev. Sci. Instrum. 29, 688-691 (1958).
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4. Durrant-Whyte, H. & Bailey, T. Simultaneous localization and mapping: part I. IEEE Robot. Autom. Mag. 13, 99-110 (2006).
5. US EPA Method 1603: Escherichia coli in water by membrane filtration using modified membrane-thermotolerant Escherichia coli agar (2014).